QuizRx LLC

Engineering | Technical Assessment

Backend & Cloud Migration Assessment

1) Node.js (NestJS) to Java Spring Boot rewrite
2) Google Cloud Platform to Amazon Web Services
Time (self-delivered) and cloud cost estimates

Document	Technical & Financial Assessment
Version	1.1
Date	2 November 2026
Systems assessed	quizrx-backend-api, quizrx-frontend
Classification	Confidential - internal use
Scope	Self-delivered; cloud provider costs only
Currency	US Dollars (USD), excluding taxes

All figures are planning estimates. Validate cloud prices with the AWS Pricing Calculator and GCP billing exports before budgeting.

1. Executive Summary

This document estimates two possible changes to the QuizRx platform: (A) rewriting the backend API from Node.js (NestJS framework) to Java Spring Boot, and (B) moving hosting from Google Cloud Platform (GCP) to Amazon Web Services (AWS). All work is assumed to be done **by one in-house engineer**, so no labour or contractor costs are included. The money figures are **only what is paid to the cloud providers**. Estimates come from a direct review of the current source code and deployment configuration.

~12 months	6-7 weeks	$80-105	$100-130
Java rewrite (one engineer, full-time)	Move to AWS (one engineer, full-time)	AWS per month today (Node.js)	AWS per month after Java rewrite

Item	Your time (solo, full-time)	AWS one-time cost	AWS monthly cost (current usage)
A. Node.js to Java Spring Boot	~252 working days (about 12-13 months)	None beyond normal hosting	About +$18-25/month vs Node.js (Java needs more memory)
B. Google Cloud to AWS	~36 working days (about 6-7 weeks)	$100-$300 (1 month on both clouds + file transfer)	$80-$105/month (Google Cloud today: about $45-$70)
A + B in sequence	~14 months total	$100-$300	$100-$130/month once on Java

Key points

1. **AWS will cost more than Google Cloud at today's traffic**, by about $35-$45 per month. The reason is fixed costs on AWS (a load balancer and always-on servers), while Cloud Run can scale to zero. AWS becomes 10-25% cheaper once traffic grows (Section 4.4).

2. **New AWS credits can cover the first year.** New accounts get up to $200 in Free Tier credits, and AWS Activate gives startups $1,000 or more. At $80-$105 per month, $1,000 covers about 10-12 months.

3. **Move to AWS first, then rewrite in Java.** The move is low-risk and takes about 1.5 months. Building the Java backend directly on AWS avoids setting up deployment pipelines twice.

4. **Keep Firebase Authentication and MongoDB Atlas.** Both work from AWS, so no user or data migration is needed. Move the Atlas cluster to an AWS region to avoid slow, billed cross-cloud traffic.

5. **The Java rewrite is the bigger commitment**: about a year of solo work with a feature freeze, and no cost saving on its own.

Immediate action required (independent of any migration)

During the review, plaintext service-account private keys (Firebase Admin and GCP) were found committed in the backend repository (file *deploy/backend_env.yaml*). Anyone with access to the repository or its history can use these keys.

Recommended: revoke and rotate both keys now, remove the file from the repository and its git history, and keep secrets only in Secret Manager or Infisical. Effort: about 0.5-1 day.

2. Current System Overview

The following facts were measured directly from the repositories and drive the estimates.

Aspect	Current state
Backend framework	NestJS 11 on Node.js 20, TypeScript
Backend size	~27,000 lines of TypeScript in ~313 source files (excluding duplicated folders)
API style	GraphQL (Apollo Server 5): 121 queries + 127 mutations = 248 operations; schema file of 1,678 lines. Plus 6 REST endpoints (file upload, Stripe webhook, AI chat streaming)
Data model	MongoDB via Mongoose: 41 collections/schemas
Business modules	Chat & AI tutor (learning, practice studio, short answers, reasoning), quizzes, question banks, mock exams, topics, extraction, payments & subscriptions (Stripe), CMS, notifications, feedback, newsletter, performance insights, knowledge graph (Neo4j, Pinecone), LLM inference logs, AI pipeline, admin, users, projects, files
Real-time features	Server-Sent Events (SSE) streaming of AI chat responses
Third-party integrations	Firebase Auth (Admin SDK), Stripe, OpenAI, Pinecone, Neo4j, Google Cloud Storage, Infisical (secrets), SMTP email (Nodemailer), Excel/CSV import
Automated tests	**None** (0 unit or integration test files)
Frontend	Next.js app consuming the GraphQL API (Apollo Client); separate admin portal
Hosting	GCP Cloud Run (region us-central1) for frontend and backend; Cloud Build CI/CD; Container Registry; Secret Manager; Cloud Storage; Cloud Logging. Legacy App Engine config also present

Items outside the assessed repositories

The backend calls external services configured as *NEST_CHAT_API*, *NEST_VECTOR_API* and *COGNITIVE_SERVICE_URL*. If these AI services are also hosted on GCP, they must move too: add about 3-5 working days and roughly $20-$40 per month on AWS for each service. The MongoDB, Neo4j and Pinecone hosting locations were not visible in the code and are assumed to be managed SaaS, billed the same on either cloud.

3. Part A - Rewriting the Backend in Java Spring Boot

A framework change of this kind is a full rewrite: no code can be reused automatically. The aim is a functionally identical backend that the existing frontend can use without changes.

3.1 Technology mapping

Current (Node.js)	Target (Java)	Notes
NestJS 11, TypeScript	Spring Boot 3.x, Java 21 (LTS)	Modules map to Spring packages and @Service beans
Apollo GraphQL (code-first)	Spring for GraphQL (schema-first)	Reuse the existing schema.gql as the contract
Mongoose	Spring Data MongoDB	41 schemas become @Document classes and repositories
Firebase guards, roles	Spring Security + Firebase Admin Java SDK	Same Firebase tokens, so no user migration
SSE streaming (chat)	Spring WebFlux / SseEmitter	Streaming LLM output needs careful testing
graphql-upload, multer	Spring multipart REST endpoint	Spring GraphQL has no built-in file-upload support; small frontend change may be needed
class-validator, zod	Jakarta Bean Validation	
Stripe, OpenAI, Pinecone, Neo4j SDKs	Official Java SDKs (Spring AI optional)	All have supported Java clients
Winston logging	SLF4J + Logback (JSON)	
Throttler	Bucket4j	
Nodemailer, xlsx, csv-parse	Spring Mail, Apache POI, Commons CSV	
Jest (unused)	JUnit 5, Testcontainers, GraphQL tests	New test suite

3.2 Time breakdown (one engineer)

Work package	Scope	Working days
Discovery & architecture	Requirements capture, Java project skeleton, conventions, security design	10
Core platform	Auth & roles, users, admin, projects, file management, API integrations, error handling, logging, email, configuration	22
GraphQL contract parity	Map all 248 operations, scalars, errors and pagination to the existing schema	10
Chat & streaming	Chat service and controller, SSE streaming, messages, threads, citations	18
Learning engine	Tutor, practice studio, short-answer grading, reasoning, orchestrator, knowledge index	22
Assessment content	Quizzes, questions, question bank, mock exams and history, topics, extraction	20
Payments	Stripe customers, subscriptions, payment methods and references, webhooks	15
Knowledge graph	Collections, files, embeddings, Pinecone and Neo4j configuration and validation	14
Inference & AI pipeline	LLM call logging (v1/v2), AI pipeline service	12

Work package	Scope	Working days
Content & engagement	CMS (pages, navigation, seeding), notifications, feedback, newsletter, performance insights	14
Configuration & seeders	Infisical, environment schema, data seeders	5
Automated testing	Contract tests against the Node.js version, integration tests, regression testing	25
DevOps	Dockerfile, CI/CD, JVM tuning, load testing	8
Parallel run & cut-over	Shadow traffic, data checks, go-live, stabilisation	15
Subtotal		**210**
Contingency (20%)	Unknowns, undocumented behaviour, rework	42
Total		**252 working days**

At about 21 working days per month, 252 days is **about 12-13 months of full-time work**. AI coding assistants can speed up the mechanical TypeScript-to-Java translation (realistically by 20-30% on the module work), but testing, debugging and cut-over time stay about the same. A realistic best case is therefore about 9-10 months. Working part-time on the rewrite while maintaining the live product roughly doubles the duration.

3.3 Effect on the AWS bill

Java services usually need 1-2 GB of memory per instance where Node.js needs 0.5-1 GB, and they start more slowly. On AWS Fargate the backend therefore needs a larger task size. Only the backend changes; the Next.js frontend stays on Node.js.

Scenario	Backend on Node.js	Backend on Java	Difference per month
Small (today)	1 task, 0.5 vCPU / 1 GB = ~$18	1 task, 1 vCPU / 2 GB = ~$36	+ ~$18
Growth (~10k active users)	2 tasks, 2 vCPU / 4 GB = ~$144	2 tasks, 2 vCPU / 6 GB = ~$157	+ ~$13 to $50
Scale (~50k active users)	6 tasks, 2 vCPU / 4 GB = ~$432	6 tasks, 2 vCPU / 6 GB = ~$470	+ ~$40 to $110

The upper end of each range allows for an extra task during traffic peaks. This increase can be removed by compiling the Java service to a GraalVM native image, which uses memory similar to Node.js.

3.4 Main risks

Risk	Impact	Mitigation
No automated tests exist today	High	Build contract tests against the live Node.js API before rewriting any module
Hidden business logic in AI and learning modules	High	Module-by-module rewrite with side-by-side output comparison
Frontend breaks because the API changes	High	Schema-first: keep schema.gql unchanged and diff it in CI
A year-long feature freeze with one engineer	High	Strangler approach: route migrated modules to Java one at a time, so new features can keep shipping
File uploads over GraphQL	Medium	Move uploads to a REST endpoint; small coordinated frontend change
Streaming AI responses (SSE)	Medium	Build a proof of concept in the first two weeks
Repository hygiene (duplicate folders such as 'modules 2', 'core 3')	Low	Clean up before starting so the scope is unambiguous

4. Part B - Migrating from Google Cloud to AWS

The application already runs in Docker containers, so the move is mostly an infrastructure change. Only one backend file (the Google Cloud Storage service) needs code changes.

4.1 Service mapping

Current GCP service	AWS equivalent	Migration notes
Cloud Run (frontend + backend)	Amazon ECS Express Mode on Fargate + Application Load Balancer	AWS App Runner closed to new customers on 30 Apr 2026; ECS Express Mode is AWS's recommended replacement
Cloud Build (2 pipelines)	GitHub Actions (free tier) or AWS CodeBuild	Rewrite pipelines; same Dockerfiles
Container Registry (gcr.io)	Amazon ECR	
Secret Manager	AWS Secrets Manager (or keep Infisical)	Group secrets into a few JSON secrets to limit cost ($0.40 per secret per month)
Cloud Storage	Amazon S3	Code change in gcp.storage.service.ts; copy files with rclone or AWS DataSync
Cloud Logging / Monitoring	Amazon CloudWatch	Dashboards and alerts must be recreated
Firebase Authentication	Keep Firebase (recommended)	Free up to 50,000 monthly users either way. Moving to Amazon Cognito would add 10-15 days and force password resets
MongoDB (assumed Atlas)	Keep Atlas; move cluster to an AWS region	Atlas supports live migration between clouds at no extra charge
DNS / SSL	Route 53 + AWS Certificate Manager (certificates are free)	Update Stripe webhook URLs, Firebase authorised domains, CORS

4.2 Time breakdown (one engineer)

Activity	Working days
AWS account setup (organisation, IAM, VPC, budgets and alerts)	3
Infrastructure as code (Terraform): ECS, ALB, ECR, S3, Secrets Manager, CloudWatch	6
CI/CD pipelines for frontend, backend and admin portal	4
Code changes: storage service to S3, credentials, health checks	3
Secrets migration and rotation	1
Data migration: Cloud Storage to S3	2
MongoDB Atlas move to an AWS region (if applicable)	2
Domains, SSL, webhooks, CORS, Firebase domains	2
Staging validation and load testing	4
Production cut-over, parallel run, GCP shutdown	3
Subtotal	**30**
Contingency (20%)	6
Total	**36 working days**

Duration: about **6-7 weeks full-time** for one engineer. Downtime at cut-over can be kept below 15 minutes by switching traffic gradually with weighted DNS routing.

4.3 One-time AWS / cloud costs

Cost item	Estimate (USD)
Running both clouds in parallel for about 1 month (AWS staging + production while GCP is still live)	$80 - $110
Transferring stored files out of Google Cloud (about $0.12 per GB; 100 GB = $12, 1 TB = about $120). Upload into S3 is free	$10 - $150
MongoDB Atlas move to an AWS region	$0 (live migration is included)
SSL certificates (AWS Certificate Manager)	$0
Optional: AWS Developer support plan during the move	$29 per month
Total one-time	$90 - $290
Offsets: up to $200 Free Tier credits for new AWS accounts; AWS Activate startup credits ($1,000 for self-funded startups, more through accelerators)	Can cover all of the above

4.4 Monthly hosting cost comparison

No GCP billing export was available, so three usage scenarios are modelled from the current configuration (two containerised services; us-central1 on GCP compared with us-east-1 on AWS; on-demand list prices; backend on Node.js). MongoDB Atlas, Firebase, OpenAI, Pinecone, Neo4j and Stripe cost the same on either cloud and are excluded.

Scenario	Assumptions	Google Cloud / month	AWS / month	AWS with 1-yr Savings Plan
Small (today)	2 services, about 0.5-1 vCPU each, mostly idle; 50 GB files; 100 GB traffic	$45 - $70	$80 - $105	$75 - $100
Growth (~10k active users)	Backend 2 x 2 vCPU, frontend 2 x 1 vCPU running 24/7; 500 GB files; 1 TB traffic	$450 - $550	$390 - $450	$320 - $370
Scale (~50k active users)	Backend 6 x 2 vCPU, frontend 4 x 1 vCPU; 2 TB files; 5 TB traffic	$1,500 - $1,700	$1,200 - $1,350	$1,000 - $1,150

Why AWS costs more at small scale: Cloud Run can scale to zero and has a free tier. On AWS, the load balancer (about $20-25 per month), public IPv4 addresses (about $3.65 each per month) and at least one always-on task per service create a fixed base cost. **Why AWS costs less at larger scale:** Fargate compute is cheaper per vCPU-hour (about $0.040) than always-on Cloud Run (about $0.065), and the first 100 GB of outbound traffic each month is free.

4.5 AWS monthly bill, service by service

AWS service	Small (today)	Growth	Scale
Fargate - backend tasks	$18	$144	$432
Fargate - frontend tasks	$18	$72	$144
Application Load Balancer (shared)	$22	$30	$50
Public IPv4 addresses ($3.65 each)	$15	$22	$44
S3 file storage ($0.023 per GB)	$1	$12	$47
Data transfer out (first 100 GB free, then $0.09 per GB)	$0	$83	$452
CloudWatch logs and metrics	$4	$20	$50
ECR, Secrets Manager, Route 53	$3	$5	$8
CI/CD (GitHub Actions free tier or CodeBuild)	$0 - $5	$5	$10
Total (Node.js backend)	**~$81 - $86**	**~$393**	**~$1,237**

Prices: AWS us-east-1 on-demand list prices, November 2026 (Fargate $0.04048 per vCPU-hour and $0.004445 per GB-hour; ALB $0.0225 per hour plus usage). Add the Java difference from Section 3.3 after the rewrite.

4.6 Ways to lower the AWS bill

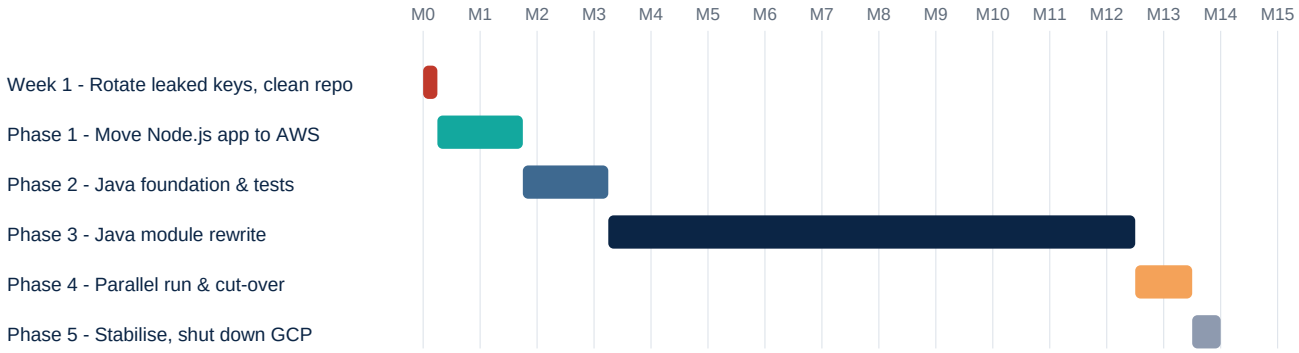
- **Use ARM (Graviton) Fargate tasks**: about 20% cheaper compute; Node.js and Java both run on ARM without code changes.
- **Put CloudFront in front of the site**: its always-free tier includes 1 TB of outbound traffic per month, which removes most data-transfer cost in the Growth scenario (about -$80 per month).
- **Commit to a 1-year Compute Savings Plan** once usage is stable: about 15-20% off Fargate with no upfront payment.
- **Set AWS Budgets alerts on day one** (free) so unexpected costs are caught within hours.
- **Keep log retention short** (for example 14-30 days) in CloudWatch.

4.7 Main risks

Risk	Mitigation
Cross-cloud latency if MongoDB stays on GCP	Move the Atlas cluster to the same AWS region before cut-over
Missed hard-coded GCP dependencies (credentials, gcr.io paths)	Search the code for GCP SDKs and credentials first; staging soak test
Webhook and OAuth breakage after the domain moves	Checklist covering Stripe, Firebase authorised domains and CORS origins
Unplanned cost growth on AWS	AWS Budgets alerts from day one; right-size tasks after 30 days

5. Recommended Plan (one engineer)

If both changes go ahead, the recommended order below takes about **14 months** of full-time work. The AWS move comes first because it is quick and low-risk, and the Java backend can then be deployed straight onto the new AWS setup.



First-year AWS budget	Small (today)	Growth	Scale
Months 1-12 on AWS, Node.js backend	$960 - $1,260	$4,700 - $5,400	$14,400 - $16,200
One-time move costs	$90 - $290	$90 - $290	$90 - $290
Minus AWS credits (example: $1,200)	- $1,200	- $1,200	- $1,200
Estimated year-1 AWS spend	$0 - $350	$3,600 - $4,500	$13,300 - $15,300
Same year on Google Cloud (for comparison)	$540 - $840	$5,400 - $6,600	$18,000 - $20,400

Credits depend on eligibility and are not guaranteed. From year 2, use the monthly figures in Section 4.4 plus the Java difference in Section 3.3.

6. Assumptions & Exclusions

- All work is done by one in-house engineer; no salaries, contractor or agency costs are included.
- One working day is 8 hours; about 21 working days per month.
- Estimates are from a static code review on 2 November 2026; no runtime traffic or billing data was analysed.
- Functional scope is fixed. New features built during the rewrite are not included.
- The frontend and admin portal are not rewritten; only minor changes (such as file upload) are included.
- MongoDB, Neo4j, Pinecone, OpenAI, Firebase and Stripe costs are the same on either cloud and are excluded.
- External AI services (NEST_CHAT_API, NEST_VECTOR_API, cognitive service) are excluded unless noted.
- Cloud prices are public on-demand list prices, excluding tax, and can change. Confirm them with the AWS Pricing Calculator and your GCP billing reports.
- Accuracy: about -15% / +30% for time and -20% / +40% for monthly cloud costs, because real traffic data was not available.